Revised Dynamic Emotion Simulation Benchmark

May 2, 2026

1 Overview

This brief report summarizes the revised first-simulation benchmark comparing DynEGA, GraphicalVAR, and GIMME on simulated dynamic emotion panels. The revised benchmark uses the old first-simulation design logic but replaces the original simulation scenarios with the newer dynamic latent-factor scenarios. DynEGA and GraphicalVAR are evaluated on the main full grid. GIMME is reported separately as a bounded feasibility comparator because full-grid GIMME estimation was not computationally tractable under the same design.

2 Simulation Benchmark Results

The main full-grid comparison contains 1,920 benchmark rows per method for DynEGA and GraphicalVAR. GIMME is limited to a 120-row bounded feasibility grid. Table 1 gives the standard benchmark metrics from the completed comparison run.

Table 1: Method-level benchmark overview. Exact recovery is the proportion of conditions in which the recovered dimensionality matched the generating structure. Runtime is mean wall-clock time per benchmark row.

Method	Scope	Runs	Success	Timeout	Error	Exact	Cor.	NRMSE	Runtime
DynEGA	Full grid	1920	1.000	0	0	0.974	0.654	0.867	3.6 s
GraphicalVAR	Full grid	1920	0.975	48	0	0.873	0.649	0.865	191.4 s
GIMME	Bounded	120	0.175	96	3	0.158	0.614	0.944	364.3 s

DynEGA is the most robust method in this benchmark. It completed all full-grid conditions, recovered the correct number of dimensions in 97.4% of rows, and had the highest score assignment accuracy. GraphicalVAR produced similar score correlations and slightly lower NRMSE, but this came at much higher runtime and with 48 full-grid timeouts. GIMME recovered dimensionality reasonably often only conditional on completion, but its all-row performance was dominated by timeouts and errors.

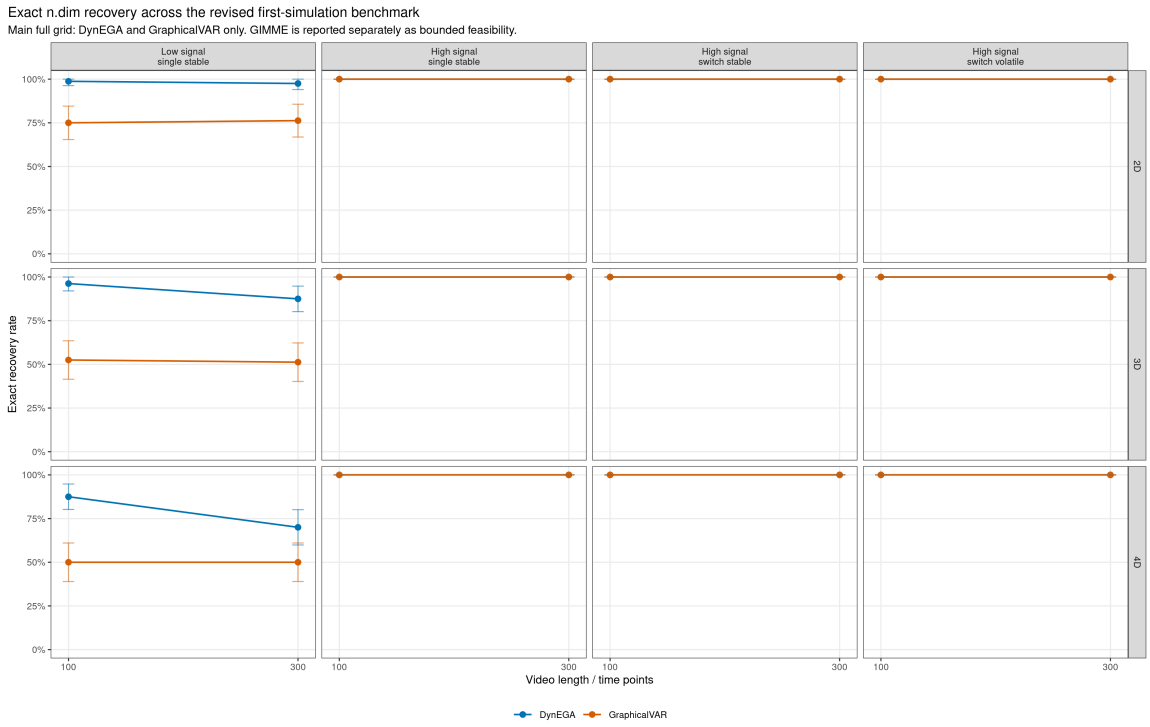


Figure 1: Exact dimensional recovery by time-series length.

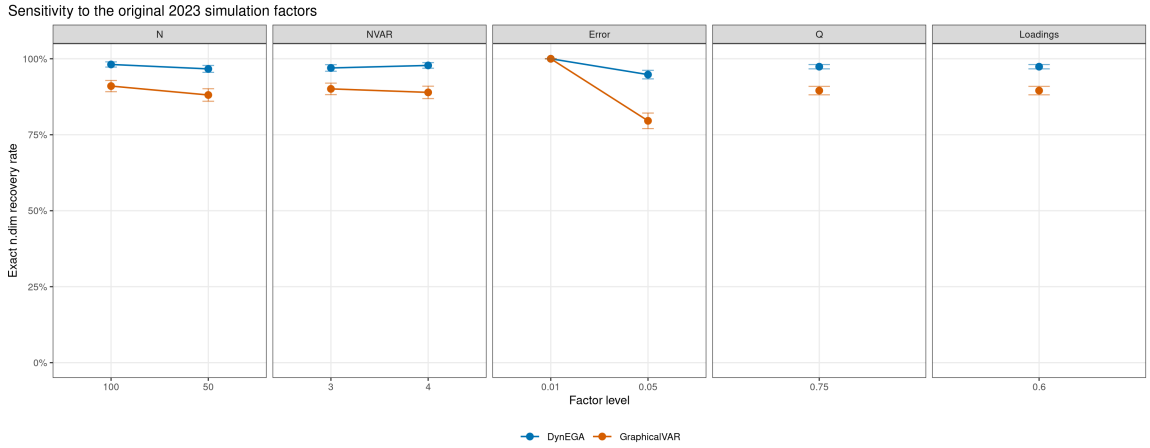


Figure 2: Sensitivity of recovery to the first-simulation design factors.

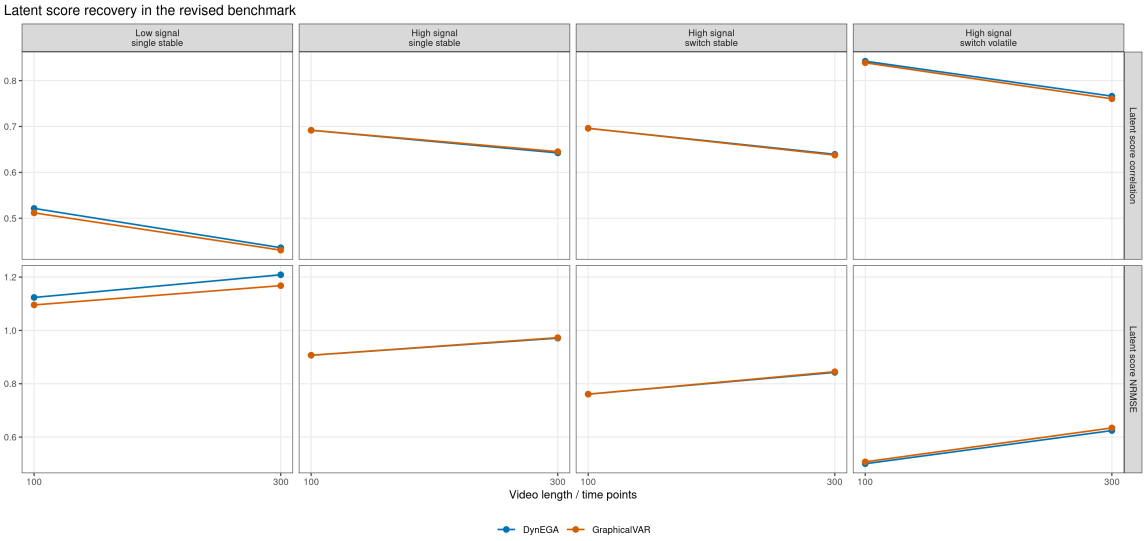


Figure 3: Recovery of latent score structure by method and scenario.

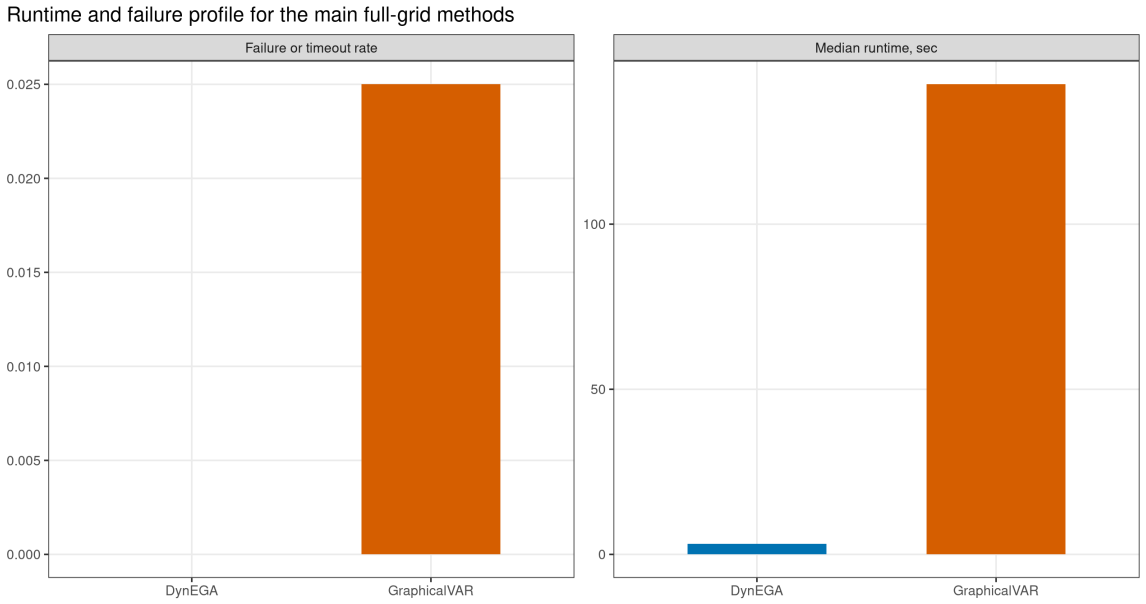


Figure 4: Runtime and failure accounting.

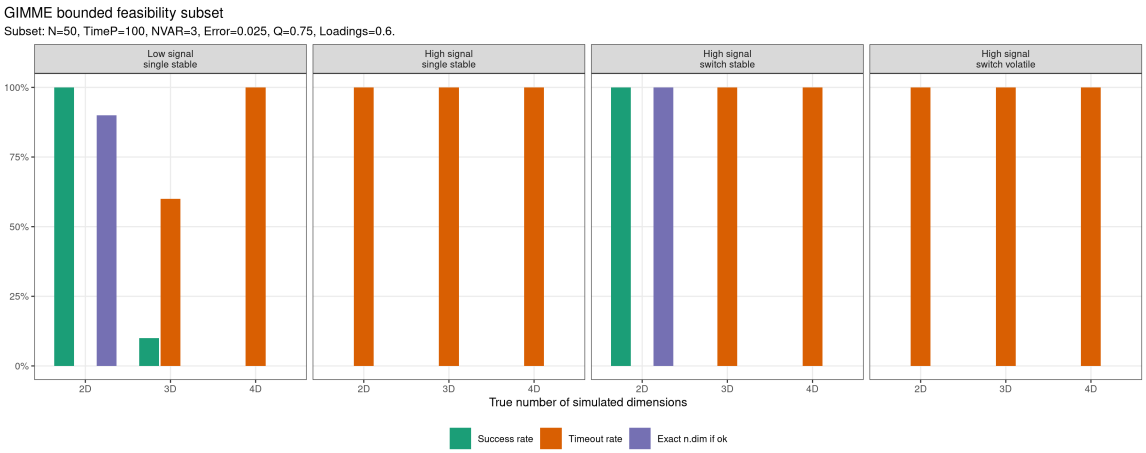


Figure 5: Bounded GIMME feasibility results. GIMME is shown as a feasibility analysis rather than a full-grid method because the same full design produced unacceptable timeout rates.

3 Empirical-Realism Check Against OAI-Large Time Series

The second check asks whether the simulated panels have time-series properties similar to observed OAI-large emotion-score series. The comparison uses the MAFW OAI-large sample and the heterogeneous DLO synthetic panel. The current similarity overview retained 86 observed base clips after empirical quality control and compared them with synthetic panels collapsed to the same base-clip level. The selected heterogeneous simulation used two latent factors with $\eta = (-0.05, -0.15)$, $\zeta = (0.08, 0.12)$, $\sigma_q = 0.25$, and $dt = 0.5$.

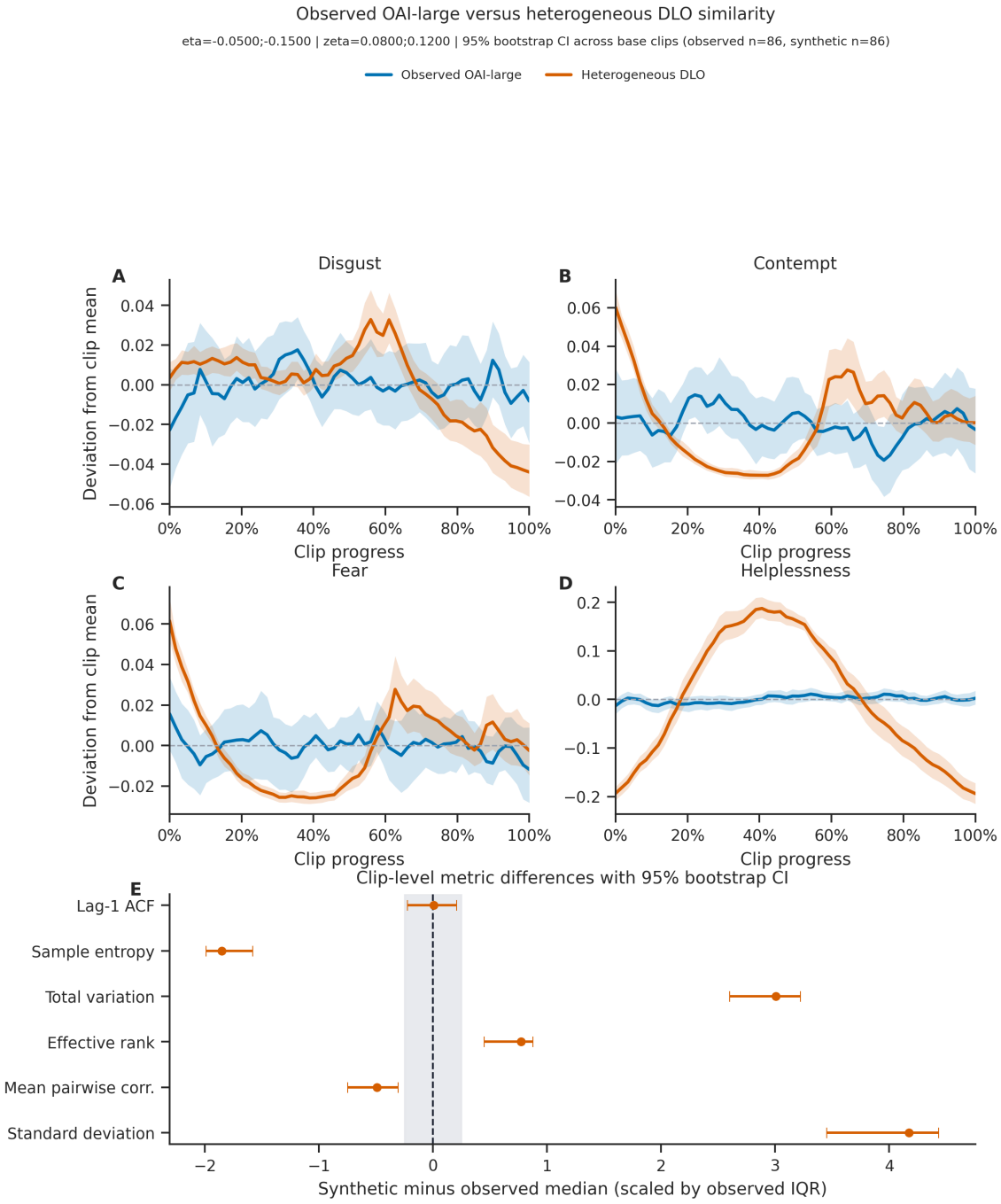


Figure 6: Observed OAI-large trajectories versus heterogeneous DLO simulations. The bottom panel shows standardized median differences in raw time-series metrics, scaled by the observed interquartile range.

The entropy-oriented summary supports the basic empirical-realism claim: the simulation is close to OAI-large on sample entropy, especially in the raw series and reasonably in the latent score series. Other variability measures show larger departures, especially standard deviation and total variation, so the claim should be framed as partial realism rather than full distributional equivalence.

Table 2: Median variability properties for OAI-large empirical series and DLO simulations. Values are medians from the 99-video OAI-large variability calibration artifact.

Family	Metric	Empirical	DLO	Difference
Raw	Sample entropy	0.410	0.421	0.011
Raw	Zero-crossing rate	0.225	0.280	0.055
Raw	Lag-1 ACF	0.542	0.408	-0.134
Raw	Standard deviation	0.020	0.090	0.070
Raw	Total variation	0.011	0.061	0.050
Latent	Sample entropy	0.427	0.392	-0.035
Latent	Zero-crossing rate	0.214	0.167	-0.048
Latent	Lag-1 ACF	0.567	0.776	0.209
Latent	Standard deviation	0.321	1.570	1.249
Latent	Total variation	0.179	0.773	0.594

4 Interpretation

The revised comparison supports three conclusions. First, DynEGA is the best default method for this design because it combines high dimensional recovery, stable completion, and low runtime. Second, GraphicalVAR is a useful structural comparator, but it is much more expensive and begins to fail in higher-dimensional or more volatile scenarios. Third, GIMME should be presented as a bounded feasibility sensitivity analysis rather than a primary benchmark competitor.

The empirical-realism checks indicate that the new DLO simulation can reproduce some important OAI-large time-series properties, particularly entropy-like irregularity measures. However, the synthetic panels still show larger movement amplitudes and stronger latent smoothness than the empirical series. This is acceptable for a method-recovery simulation, but it should be stated explicitly: the generator is calibrated for plausible dynamic structure and entropy-like behavior, not for exact reproduction of every marginal variability statistic.